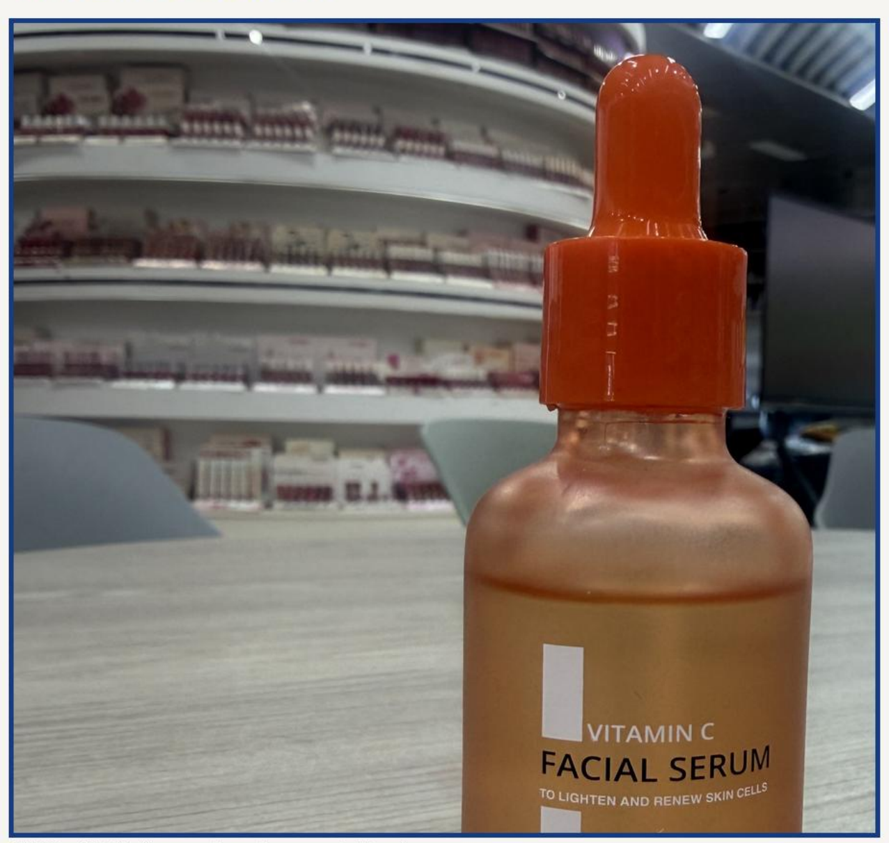
# P1 DROPPER SERUM BOTTLE - OWNER VISUAL REVIEW BOARD (V4)

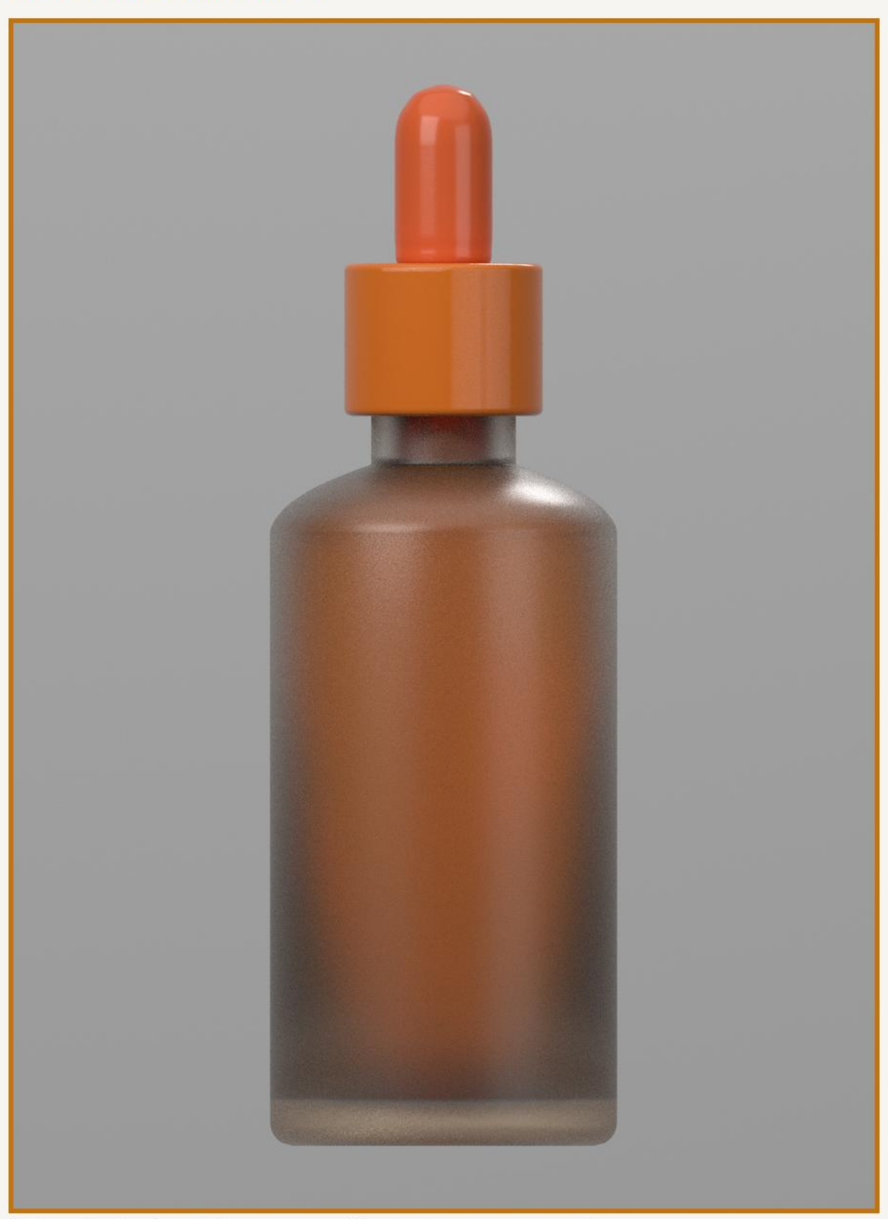
V4 REVISION - V1, V2 and V3 all owner-REJECTED and preserved unchanged I owner approval: PENDING_VISUAL_REVIEW I GATE_2: BLOCKED I manufacturing authority: NOT_ESTABLISHED
sections: OWNER PHOTOGRAPHY I OWNER-MEASURED DIMENSIONS I OWNER-CONFIRMED MATERIALS I ONLINE-ANALOGOUS EVIDENCE I CALCULATED I VISUALLY ESTIMATED I NEUTRAL GEOMETRY I MATERIAL VALIDATION I VISUALIZATION-ONLY INNER GLASS I OMITTED DETAILS
photographs are perspective + uncalibrated - NO calibrated overlay accuracy claimed I THREAD_GEOMETRY_NOT_MODELLED I masks = unlit alpha authority I judged against the real photographs, never against V3

OWNER PHOTOGRAPH



IMG_3120 (crop, lossless rotation)

MATERIAL VALIDATION



V4 material front (transparent frosted orange glass)

MATERIAL VALIDATION

V4 material three-quarter

NEUTRAL GEOMETRY VALIDATION

V4 neutral front (clay)

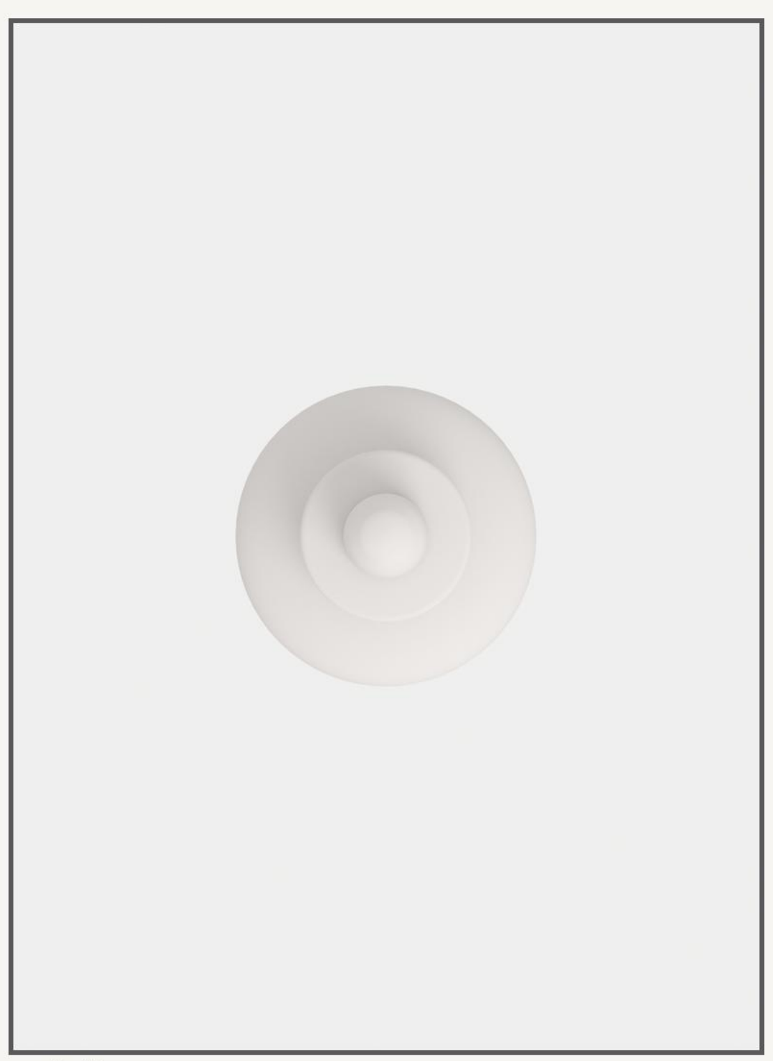
OWNER PHOTOGRAPH

IMG_3123 dropper (pipette hidden inte

NEUTRAL GEOMETRY VALIDATION

neutral side

NEUTRAL GEOMETRY VALIDATION



neutral top

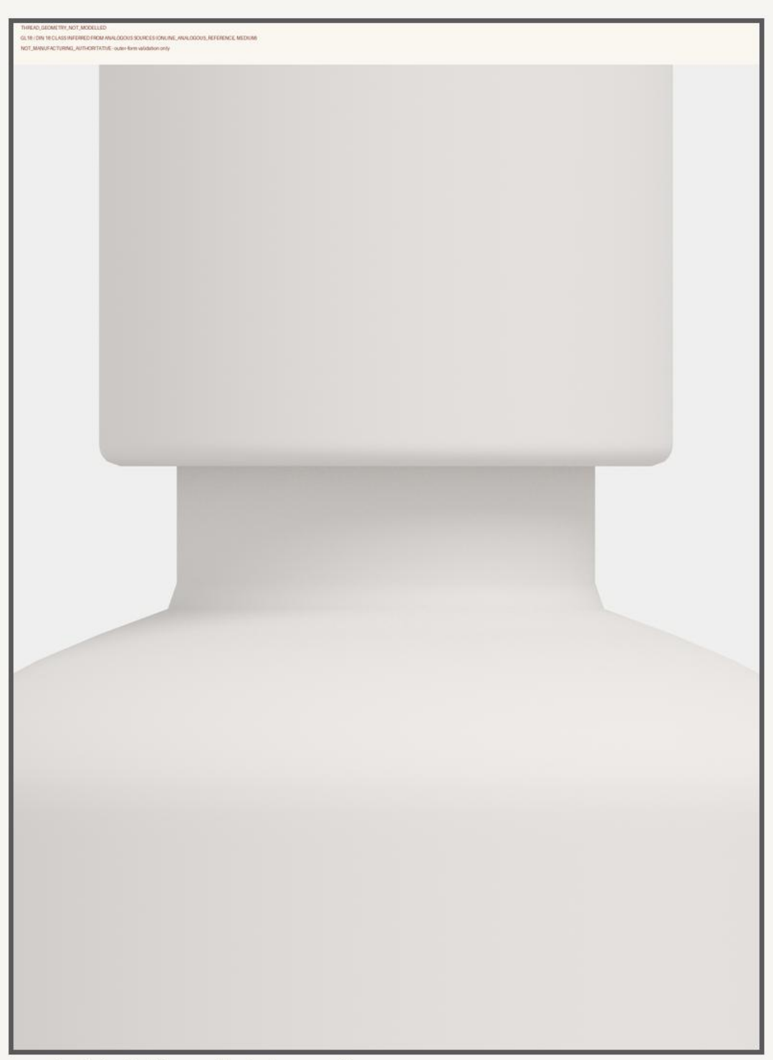
NEUTRAL GEOMETRY VALIDATION

neutral three-quarter

NEUTRAL GEOMETRY VALIDATION

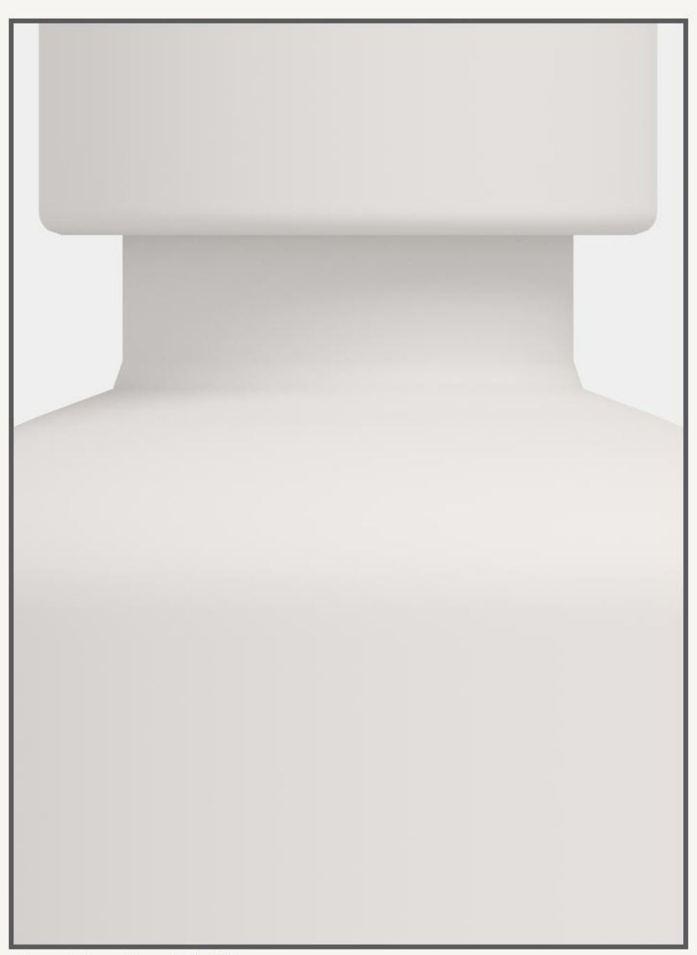
closure detail

NEUTRAL GEOMETRY VALIDATION



neck detail (thread statement)

NEUTRAL GEOMETRY VALIDATION



shoulder detail (V4)

NEUTRAL GEOMETRY VALIDATION

bulb detail (V4)

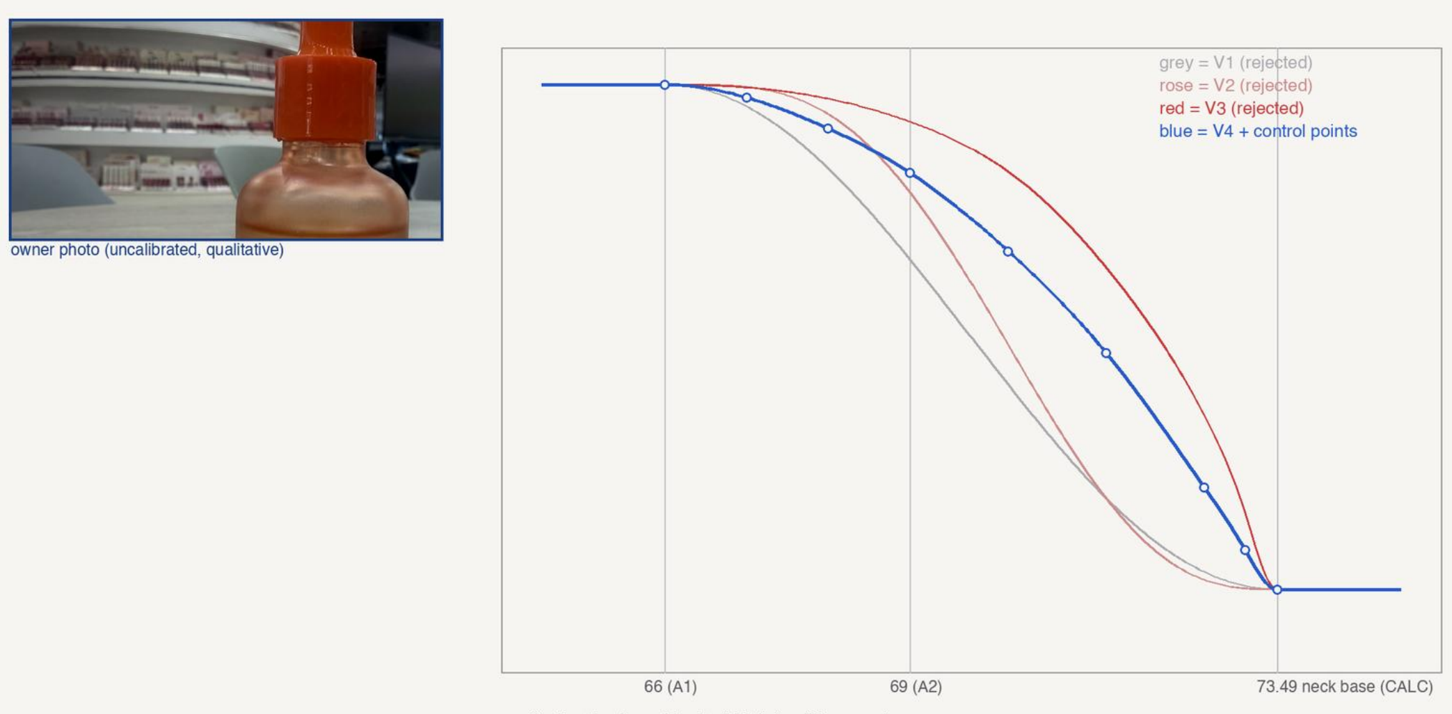
NEUTRAL GEOMETRY VALIDATION

base detail (V4)

VISUALLY ESTIMATED

## P1 SHOULDER PROFILE DIAGNOSTIC - V1 / V2 / V3 / V4

V1-V3 owner-rejected; V4 = authored clamped cubic spline, narrowing departs EARLIER + progressive curvature + soft neck root (VISUAL_ESTIMATE, LOW/MEDIUM) - NOT_MANUFACTURING_AUTHORITATIVE



owner photo (uncalibrated, qualitative)

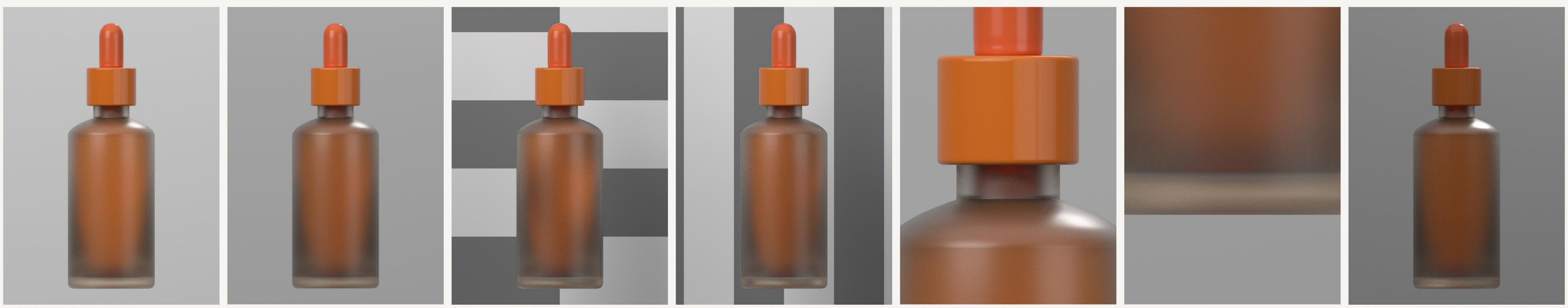
continuity at authored knots (V4, finite differences):
z= 66.00 r= 18.70 dr/dz= -0.003 d2r/dz2= -0.33
z= 69.00 r= 16.80 dr/dz= -1.160 d2r/dz2= -0.53
z= 71.40 r= 12.90 dr/dz= -2.153 d2r/dz2= -0.69
z= 73.10 r= 8.65 dr/dz= -3.056 d2r/dz2= -1.99
z= 73.49 r= 7.80 dr/dz= -0.088 d2r/dz2= 8.76
clamped tangents: dr/dz = 0 at the wall (66) and at the neck base (73.49) - no ridge, no kink

shoulder profile diagnostic (V1-V4 + V4 control points)

MATERIAL VALIDATION

## MATERIAL DIAGNOSTIC - TRANSPARENT MATTE ORANGE GLASS

proves: transparent + orange-tinted + frosted exterior + background visible through the body (diffused) + base optically denser than the neck I hollow glass is VISUALIZATION_ONLY_GEOMETRY



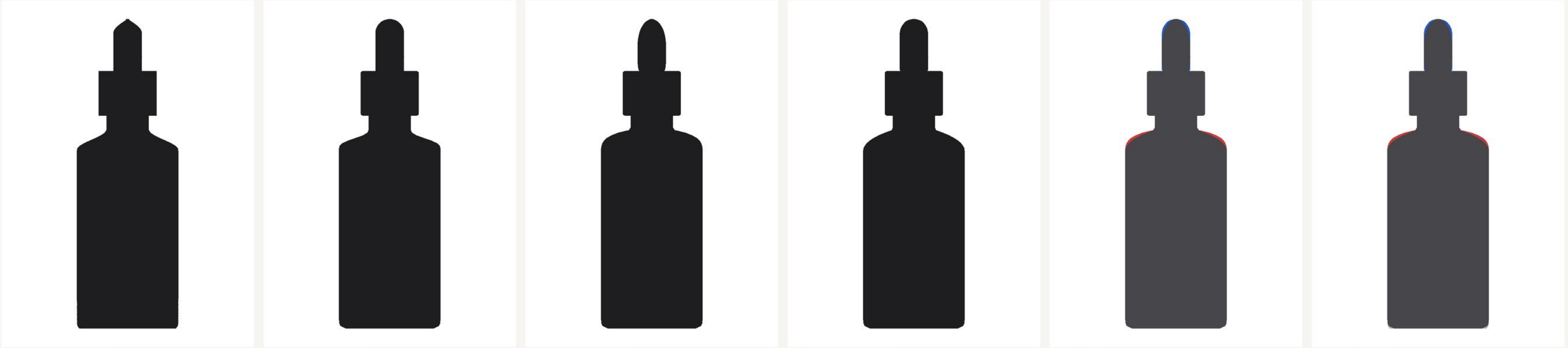
1. white background
2. mid-grey background
3. B/W calibration card behind
6. stripe object behind the bottle
4. neck + shoulder transmission
5. glass base close-up
7. side-lit edge transmission

glass-transmission diagnostic - transparent frosted orange glass (VISUALIZATION-ONLY inner glass)

CALCULATED

## P1 V1 / V2 / V3 / V4 SILHOUETTES - deterministic mask authority (unlit alpha pass; NO thresholded shaded RGB; same pinned camera)

V4 changes: shoulder narrows EARLIER with progressive curvature and a soft neck root (no swelling, no pinch) + bulb with near-vertical sides I unchanged: body wall, base, collar bounds, every accepted owner dimension I overlay: red=V3 only, blue=V4 only, dark=agreement



V1 - OWNER REJECTED
V2 - OWNER REJECTED
V3 - OWNER REJECTED
V4 - CURRENT REVISION
V3/V4 overlay
V1/V3/V4 combined

V1/V2/V3/V4 silhouettes - deterministic mask authority

### DIMENSION RECONCILIATION (mask authority)

| | | |
|---|---|---|
| assembled height | 114.3 OWNER | 114.28 mask |
| body diameter | 37.4 OWNER | 37.47 mask |
| front centred | - | 0.0006 offset |
| closure centred | - | 0.0006 offset |
| internal holes | 0 required | 0 |
| mask determinism | required | True |

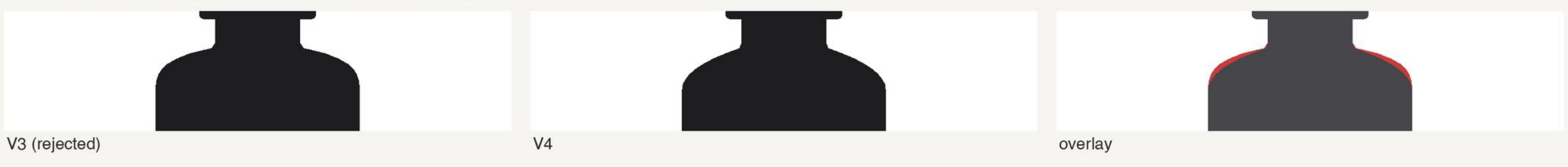
### OMITTED MANUFACTURING DETAILS

GL18-class thread (stamped on views) I base embossing
50ml/46 I label artwork I pipette in renders
inner glass = VISUALIZATION_ONLY (never manufacturing)

CALCULATED

## V3 vs V4 SHOULDER (close-up)

mask authority (unlit alpha) I red=V3 only, blue=V4 only, dark=agreement



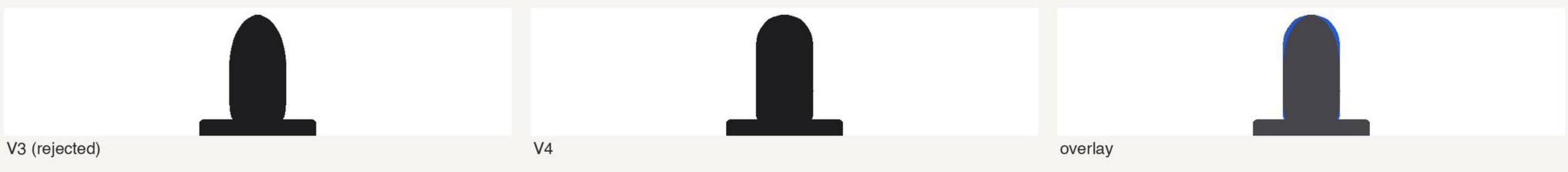
V3 (rejected)
V4
overlay

V3 vs V4 shoulder close-up

CALCULATED

## V3 vs V4 BULB (close-up)

mask authority (unlit alpha) I red=V3 only, blue=V4 only, dark=agreement



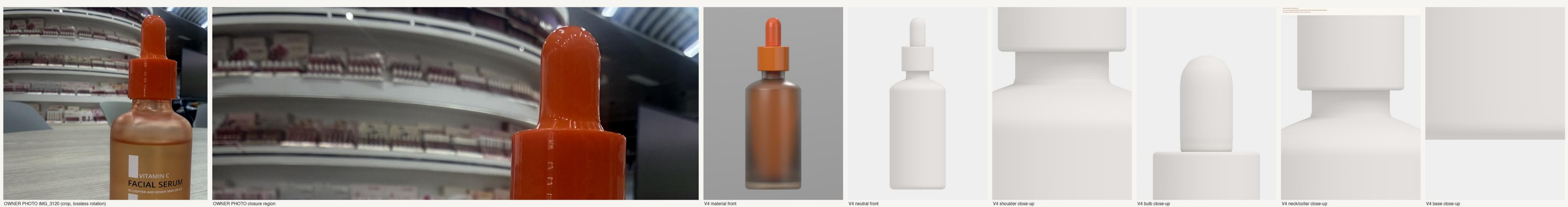
V3 (rejected)
V4
overlay

V3 vs V4 bulb close-up

OWNER PHOTOGRAPH

## P1 OWNER PHOTO vs V4 - QUALITATIVE ONLY

OWNER PHOTOGRAPH IS PERSPECTIVE AND UNCALIBRATED I NO ACCURATE PHOTOGRAPHIC OVERLAY IS CLAIMED I COMPARISON IS QUALITATIVE ONLY I photos never stretched/warped



OWNER PHOTO IMG_3120 (crop, lossless rotation)
OWNER PHOTO closure region
V4 material front
V4 neutral front
V4 shoulder close-up
V4 bulb close-up
V4 neck/collar close-up
V4 base close-up

owner photo vs V4 (qualitative only - uncalibrated)

evidence: docs/visual_quality/rbv/p1_pilot/2026-08-04_v4 I V1-V3 preserved I contract p1_provisional_geometry/1.0 revision V4
visual review + pipeline validation only - NEVER supplier/manufacturing geometry